Stealthy Computational Delay Attacks on Control Systems

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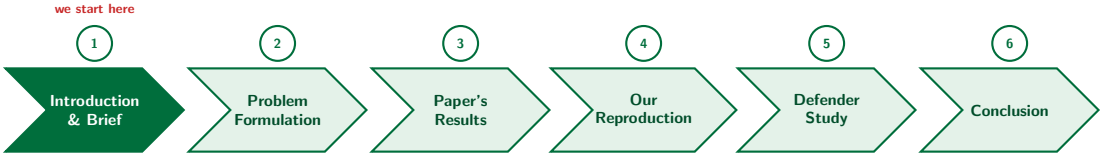
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Source: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025 · DOI 10.1145/3716550.3722013

Presentation plan



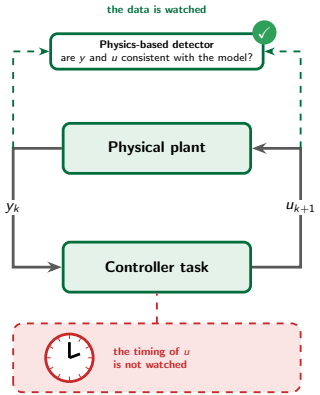
Source: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025 · DOI 10.1145/3716550.3722013

Introduction: why timing is a security problem

- A **cyber-physical system** couples computation with a physical process: power grids, factories, vehicles, medical devices.
- Its controller is a **periodic task**: each period it reads a sensor measurement y and must deliver a new command u before the **deadline**.
- Many attacks tamper with data. **Stuxnet (2010)** manipulated PLC commands while replaying normal-looking sensor data.
- So defences learned to check the data. **Physics-based detectors** ask whether y and u agree with the plant model.

What detectors don't check

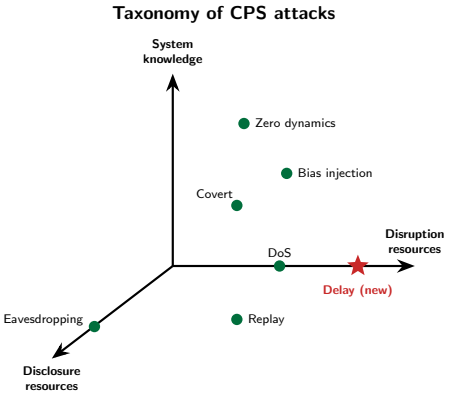
A detector checking the plant against the applied u can miss changes to *when* commands arrive. A controller- or timing-aware detector can still detect them.



Schematic: y is the sensor measurement; u is the control command.

Source: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025 · DOI 10.1145/3716550.3722013

Introduction: where delay attacks fit



Schematic placement after Teixeira et al.
A. Teixeira, I. Shames, H. Sandberg & K. H. Johansson, *Automatica* 51 (2015), 135–148.

Positions are qualitative. Execution needs disruption capability; worst-case planning uses the model.

	False data injection	Denial of service	Computational delay
Changes	sensor or actuator values	availability	timing of the control job
Reads data?	attack-dependent	no	no
Needs plant model?	attack-dependent	not necessarily	to plan the worst case
Appearance	model mismatch or stealthy forgery	often conspicuous	normal-looking timing jitter
Plant-residual detector?	often, not always	may expose disruption	usually not, if it checks the applied u

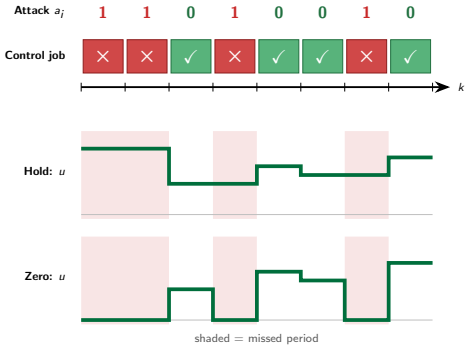
Why timing monitoring matters

Normal operation can include deadline misses. Selected extra misses can pass the timing tests even when physical impact increases.

Source: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025 · DOI 10.1145/3716550.3722013 · Taxonomy: Teixeira et al., *Automatica* 2015; paper Fig. 1

Brief of the paper: the attack idea

- **CPU starvation:** the attacker can pre-empt or block the controller, through sufficient priority or a shared resource, so a job misses its deadline.
- Each period the attacker decides: **0 = let it run**, **1 = force a miss**.



Binary misses and the two actuator responses; illustrative commands.

What happens on a miss

- No fresh command is available; z and the stored measurement remain frozen.
- **Hold:** keep the old u. **Zero:** set $u = 0$.
- The delayed computation eventually completes using the **stored** measurement (paper Eq. 3).

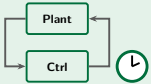
What the attacker needs

- **To act:** a task able to pre-empt or block the control job.
- **No** sensor data, **no** network access, **no** forged signals.
- **To plan the worst case:** the plant/controller model, initial state and detector settings.

Source: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025 · DOI 10.1145/3716550.3722013

The study at a glance

1



The setting

Section 3.1

A linear plant and a periodic controller that shares its CPU.

2



The attack

Sections 4.1–4.2

Starve chosen jobs so they miss the deadline; u is held or zeroed.

3



The detectors

Sections 3.2, 4.4

A watchdog, a binomial window test and a geometric (χ^2) test.

4



The optimisation

Section 4.3

Pick the miss pattern that does the most damage yet passes every test.

5



The experiments

Section 5

A stable quadruple tank (simulated) and an unstable Furuta pendulum (real hardware).

Research question

How can an attacker maximise physical disruption while avoiding timing-based detection? The paper computes the worst attack **over a finite horizon**, under its model and detector constraints.

Source: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025 · DOI 10.1145/3716550.3722013

Key takeaways

1. **Timing is a security property**
   Deadline misses are usually treated as a performance issue. This paper shows they are an attack surface that needs its own monitoring.

2. **Stealth is a statistical idea**
   "Undetected" means every active test still fails to reject normal behaviour at level α. A smart attacker plans right up to that edge.

3. **The attacker needs very little**
   A task with sufficient priority or blocking capability can delay the controller. No forged signals or network access are required.

4. **Defence must span layers**
   Scheduling, detection and control design all matter. What the actuator does on a miss can decide whether the plant survives.

5. **The finite-horizon worst case is computable**
   Optimisation gives designers a damage bound to test their defences against, instead of guessing.

Timing and control

Real-time scheduling is part of security. The next part formalises this model step by step.

Source: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025 · DOI 10.1145/3716550.3722013

PRESENTATION PART 2

Problem Formulation

Paper Sections 3–4

System model, attack optimisation and timing-based detection

Presented by

Swarali Nilesh Savale

Source: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025 · DOI 10.1145/3716550.3722013

Symbol glossary: plant, controller and timing

Symbol	Meaning
k	period/sample index (starts at 0)
$h,\ T_s$	control period / sample time
x_k	physical plant state
y_k	plant output / sensor measurement
u_k	input applied during period k
u_{k+1}	command for the next period
z_k	controller's internal state
$\tilde{y}_k$	stored measurement used by the delayed computation

Symbol	Meaning
A, B, C, D	discrete-time plant matrices
F, G, H, K	controller matrices
ψ_k	augmented state $[x_k; \tilde{y}_k; z_k; u_k]$
Φ_h, Φ_m	hit/miss evolution matrices
Δ	actuator-memory block: identity (HOLD), zero (ZERO)
a_k	binary decision: 0 hit, 1 miss
ℓ	attack horizon in periods
θ	Furuta pendulum angle (rad); angle-squared cost in rad^2

Source: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025 · DOI 10.1145/3716550.3722013 · Paper notation; Eqs. 1–7

Symbol glossary: detectors, costs and project links

Symbol	Meaning
$\bar{a}$	maximum allowed consecutive misses
n_w	binomial-window length
$\hat{p}_m$	estimated nominal miss probability
α	per-test significance level
$\mathcal{B}_{1-\alpha}$	binomial critical miss count
$m_r,\ p_r$	observed count / probability of r misses between hits
$N = \sum_r m_r$	number of counted hit-to-hit intervals
$E_r = Np_r$	expected count in category r
S_{χ^2}	textbook geometric (χ^2) statistic
$T = NS_{\chi^2}$	equivalent scaled statistic used by REP

Symbol	Meaning
Q	fixed positive-semidefinite output weight
$J_{\mathrm{attack}}, J_{\mathrm{nominal}}$	terminal cost with / without the attack
$\omega = J_{\mathrm{attack}} - J_{\mathrm{nominal}}$	extra terminal cost for a fixed horizon
ω^*	largest feasible ω for the configuration
$d,\ \mathcal{L}$	defender choice / tested horizons
FA, φ	session false-alarm probability / allowed limit
H_{sess}	declared session length for the FA experiment

Editable project-link placeholders

GitHub: your-team/cps-delay-attack

Google Drive: project-folder-placeholder

These are dummy links, supplied as editable placeholders. Replace the two URL macros before sharing.

Source: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025 · DOI 10.1145/3716550.3722013

Part 2: the model

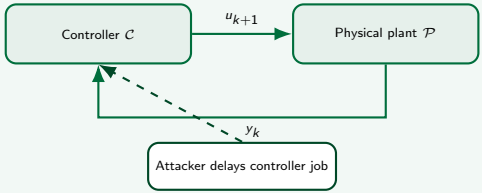
Attack scenario

- A real-time controller must finish each control computation before its deadline.
- An attacker can cause **CPU starvation or execution delays**, making a control job miss its deadline.
- The attacker need not alter the sensor measurement or plant equations. The physical plant may be harmed because a fresh control command is unavailable on time.

Research question

How can an attacker maximise physical disruption while avoiding timing-based detection?

Closed-loop attack path



Schematic closed-loop timing attack path.

The plant continues evolving even when the controller misses a deadline.

Modelling scope

Linear plant and controller; periodic execution; independent nominal misses with estimated probability $\hat{p}_m$. The attacker changes the hit/miss sequence.

Source: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025 · DOI 10.1145/3716550.3722013

Discrete-time plant and controller

Physical plant — Eq. (1)

$$\begin{aligned} x_{k+1} &= Ax_k + Bu_k, \\ y_k &= Cx_k + Du_k. \end{aligned}$$

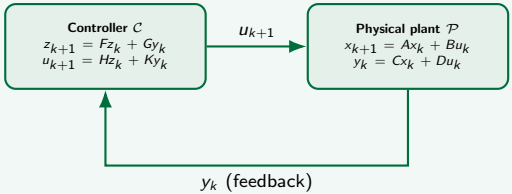
- x_k: plant state
- u_k: applied plant input
- y_k: plant output/measurement

Controller — Eq. (2)

$$\begin{aligned} z_{k+1} &= Fz_k + Gy_k, \\ u_{k+1} &= Hz_k + Ky_k. \end{aligned}$$

- z_k: controller state
- y_k: measured plant output
- u_{k+1}: command for the next period

Complete physical feedback loop



Discrete-time plant/controller feedback and the one-period command convention.

Timing convention

During period k, the plant holds u_k constant. The controller measures y_k and computes u_{k+1} for period $k+1$. The nominal control objective is to regulate the output toward zero.

Source: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025 · DOI 10.1145/3716550.3722013

Computational delay attack and switched dynamics

Augmented closed-loop state

The state includes the plant, controller, and stored information:

$$\psi_k = \begin{bmatrix} x_k \\ \tilde{y}_k \\ z_k \\ u_k \end{bmatrix}$$

$\tilde{y}_k$ is the latest stored measurement. The augmented state ψ_k captures the complete closed-loop state used by the attack model.

Binary deadline decision

$$a_k = \begin{cases} 0, & \text{deadline hit,} \\ 1, & \text{deadline miss.} \end{cases}$$

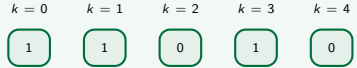
$$\psi_{k+1} = \left[(1 - a_k)\Phi_h + a_k \Phi_m\right] \psi_k$$

Φ_h and Φ_m are the hit- and miss-mode matrices (paper Eqs. 4–6; the full optimisation is Eq. 7).

What happens on a missed deadline?

- **Miss:** x evolves; z and $\tilde{y}$ stay frozen.
- **Hit:** computation completes using $\tilde{y}_k$; z and u update, and $\tilde{y}_{k+1} = y_{k+1}$ refreshes the buffer.
- The actuator either **HOLDs** the previous command or applies **ZERO**.

Example sequence: 11010



Five illustrative decisions $a_0, \ldots, a_4$. REP later stores this interior sequence as 0 11010 0, with forced boundary hits.

1 = miss; 0 = hit. The attacker chooses timing failures, not the output directly.

Source: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025 · DOI 10.1145/3716550.3722013

Hit and miss matrices in the paper

The augmented state is $\psi_k = [x_k^\top \;\; \tilde{y}_k^\top \;\; z_k^\top \;\; u_k^\top]^\top$.

Deadline hit: paper Eq. 5

$$\Phi_h = \begin{bmatrix} A & 0 & 0 & B \\ CA & DK & DH & CB \\ 0 & G & F & 0 \\ 0 & K & H & 0 \end{bmatrix}.$$

The job completes using $\tilde{y}_k$. The stored measurement then refreshes to y_{k+1}.

Deadline miss: paper Eq. 6

$$\Phi_m = \begin{bmatrix} A & 0 & 0 & B \\ 0 & I_{n_y} & 0 & 0 \\ 0 & 0 & I_{n_z} & 0 \\ 0 & 0 & 0 & \Delta \end{bmatrix}.$$

$\Delta = I_{n_u}$ for HOLD; $\Delta = 0$ for ZERO. The plant evolves while z and $\tilde{y}$ remain fixed.

Switched model: Eq. 4, written in the binary form of Eq. 7

$$\psi_{k+1} = [(1 - a_k)\Phi_h + a_k\Phi_m]\psi_k.$$

Source: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025 · DOI 10.1145/3716550.3722013 · Eqs. 3–7

From objective to effectiveness ω

Finite-horizon optimisation: paper Eq. 7

$$
\begin{aligned}
\max_{a_0,\dots,a_{\ell-1}} \quad & y_\ell^\top Q y_\ell \\
\text{subject to} \quad & a_k \in \{0,1\}, \quad \psi_0 \text{ given}, \\
& \sum_{q=j}^{j+\bar{a}} a_q \le \bar{a}, \\
& \psi_{k+1} = [(1-a_k)\Phi_h + a_k \Phi_m]\psi_k, \\
& \sum_{q=j}^{j+n_w-1} a_q \le \mathcal{B}_{1-\alpha}(n_w, \hat{p}_m), \\
& S_{\chi^2}(a) \le \chi^2_{1-\alpha}(\bar{a}).
\end{aligned}
$$

Constraints apply to each valid index/window. Short sequences use the REP boundary convention.

Gurobi searches over miss sequences. Q is a fixed design weight.

Maximised and reported quantities

$$J_{\text{attack}} = y_\ell^\top Q y_\ell, \qquad \omega = J_{\text{attack}} - J_{\text{nominal}}.$$

The same initial state gives a constant nominal cost. Maximising J_{attack} also maximises ω.

Output weighting Q

Furuta: $Q = \text{diag}(1, 0, 0)$, scoring pendulum angle.

Tank: $Q = \text{diag}(1, 1)$, scoring both regulated outputs.

Limitation

Only the terminal output counts. Earlier damage is not scored.

Source: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025 · DOI 10.1145/3716550.3722013 · Eqs. 7, 9–11; Definition 5.1

Detection I: watchdog and binomial test

Watchdog: consecutive misses

No more than $\bar{a}$ consecutive misses

- $a_k = 1$ denotes a deadline miss.
- If $\bar{a} = 2$, three consecutive misses violate the watchdog rule.
- Passing the watchdog limits runs, but not necessarily the total number of misses in a window.

Model assumption

Nominal misses are modelled as independent events with estimated probability $\hat{p}_m$. This is an assumption, not a universal property of every real-time system.

Sliding-window binomial test

Let $w_i \in \{0, 1\}$ denote the hit/miss status of job i in a window of n_w jobs, where 1 denotes a miss.

$$S_w = \sum_{i=1}^{n_w} w_i$$

Under the nominal model,

$$S_w \sim \text{Binomial}(n_w, \hat{p}_m).$$

The one-sided hypotheses are

$$H_0 : p \leq \hat{p}_m, \qquad H_1 : p > \hat{p}_m.$$

The sequence passes the binomial test when

$$\boxed{\sum_{i=1}^{n_w} w_i \leq \mathcal{B}_{1-\alpha}(n_w, \hat{p}_m)}$$

$\mathcal{B}_{1-\alpha}$ is the binomial critical count; α is the significance level.

Key distinction

Watchdog: limits *consecutive* misses. Binomial: limits the *total* in a window. Passing a test does not prove that no attack is present.

Source: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025 · DOI 10.1145/3716550.3722013

Detection II: the geometric (χ^2) test

Run-length categories

Let R be the number of consecutive misses before a hit. The final category groups the tail at the watchdog limit:

$$P(R = r) = \begin{cases} \hat{p}_m^r(1 - \hat{p}_m), & 0 \le r < \bar{a}, \\ \hat{p}_m^{\bar{a}}, & r = \bar{a}. \end{cases}$$

The categories are $r = 0, 1, \ldots, \bar{a}$; the final category groups the tail run length.

Expected and observed counts

If $N = \sum_r m_r$ counts the completed hit-to-hit intervals used by the REP test, then

$$\boxed{E_r = NP(R = r)}$$

where m_r is the observed count for category r.

Chi-square statistic

$$\boxed{S_{\chi^2} = \sum_{r=0}^{\bar{a}} \frac{(m_r - E_r)^2}{E_r}}$$

The acceptance condition is

$$\boxed{S_{\chi^2} \le \chi^2_{1-\alpha}(\bar{a})}$$

$\chi^2_{1-\alpha}(\bar{a})$ is the critical value at significance level α, using the paper's degrees of freedom. Small expected category counts weaken the χ^2 approximation; empirical calibration is needed.

Why use both statistical tests?

A sequence may satisfy the watchdog yet have an abnormal run-length pattern. The binomial test checks miss counts; the geometric (χ^2) test checks run-length frequencies.

Source: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025 · DOI 10.1145/3716550.3722013

PRESENTATION PART 3

Experimental Results

Paper Section 5: Experimental Results

Quadruple-tank (5.2) · Furuta pendulum (5.3) · Figures 2–4 and Table 1

Presented by

Madhavi Singh

Source: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025 · DOI 10.1145/3716550.3722013

Optimisation

A Mixed-Integer Quadratically-Constrained Program (MIQCP) in Gurobi (`gurobipy`), with indicator constraints for the hit/miss dynamics.

What is measured

Per configuration: the optimal attack schedule, the optimal and nominal objectives, and effectiveness $\omega = J_{\text{attack}} - J_{\text{nominal}}$.

Swept parameters

- attack horizon $\ell \in \{1, \dots, 25\}$
- watchdog threshold $\bar{a} \in \{1, \dots, 5\}$
- detection window $n_w \in \{10, 12, \dots, 20\}$
- nominal miss prob. $\hat{p}_{\text{m}} \in \{0.01, 0.03, \dots, 0.25\}$
- detector: binomial, geometric (χ^2), combined

Note: runtime depends on the grid, solver and licence. The authors estimate about 2 h for the reduced experiment; a full campaign is much larger.

Source: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025 · DOI 10.1145/3716550.3722013

Quadruple-tank process

System A: stable, non-minimum-phase, simulated (HOLD)

Setup

- Four interconnected tanks; two pumps, cross-coupled.
- Non-minimum-phase configuration (hard to control).
- LQG controller; HOLD policy on a miss; $T_s = 0.1\,\text{s}$.
- Open-loop **stable** — evaluated in simulation.
- Objective weights the two lower-tank outputs.

What the attack does here

Because the plant is stable, delay does not destabilise it — it **degrades regulation**: levels drift from setpoint and settle slowly, bounded and recoverable. ω is plant-specific: outputs, Q and scales differ. Compare shapes, not raw values, across plants.

Source: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025 · DOI 10.1145/3716550.3722013

Furuta rotary pendulum

System B: unstable, LQR, HOLD & ZERO, real hardware

Setup

- Rotary arm + pendulum; controller balances it upright.
- Open-loop **unstable** — the high-stakes case.
- LQR controller; **both** HOLD and ZERO tested; $T_s = 5\,\text{ms}$.
- Validated on **real hardware** (with a video).
- Objective weights the pendulum angle.

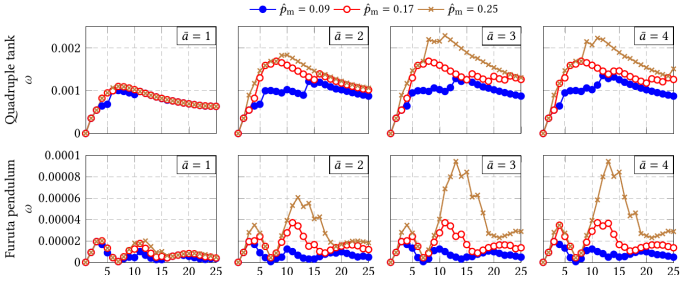
What the attack does here

A well-timed stealthy burst of misses under HOLD drives the pendulum off upright and can **topple it**. ZERO keeps it near upright for this plant and tested attack. Policy benefits are plant-dependent; see the tank replay later.

Source: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025 · DOI 10.1145/3716550.3722013

Figure 2: effectiveness vs attack length

Both plants; combined binomial and geometric (χ^2) tests; $n_w = 20$



Paper Fig. 2: tank HOLD (top), Furuta ZERO (bottom); columns $\bar{a} = 1, \ldots, 4$.

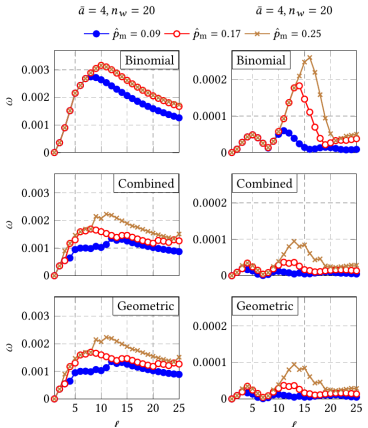
Takeaways

- Effectiveness peaks around $\ell \approx 10$–13, depending on the panel. Longer attacks are not always worse.
- From $\bar{a} \approx 3$, additional allowed consecutive misses barely change these curves.
- Tank $\sim 10^{-3}$ and Furuta $\sim 10^{-5}$ use different output costs. Compare shapes only.

Source: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025 · DOI 10.1145/3716550.3722013 · Fig. 2

Figure 3: detector comparison

Binomial, combined and geometric (χ^2) tests; $\bar{a} = 4$, $n_w = 20$



Paper Fig. 3: tank HOLD (left), Furuta ZERO (right).

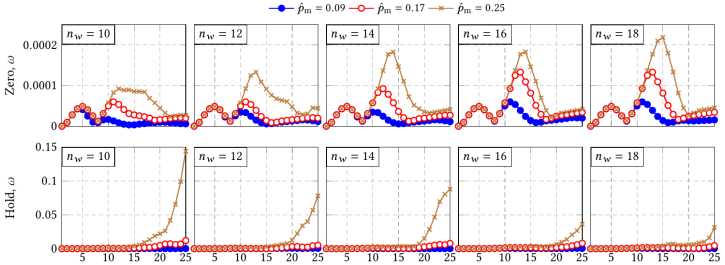
Takeaways

- Rows compare binomial, combined and geometric (χ^2) tests. The horizontal axis is ℓ; curves use $\hat{p}_m = 0.09, 0.17, 0.25$.
- In these panels the combined and geometric (χ^2) results coincide: the geometric (χ^2) constraint is binding.
- Our planned defender study includes all three detector choices. A fair ranking also needs an empirical session false-alarm comparison.

Source: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025 · DOI 10.1145/3716550.3722013 · Fig. 3

Figure 4: ZERO vs HOLD (Furuta)

Window $n_w = 10, 12, \ldots, 18$; $\bar{a} = 4$; binomial test



Paper Fig. 4: ZERO (top) and HOLD (bottom); note the different vertical scales.

Takeaways

- ZERO reaches about 2×10^{-4}; HOLD reaches about 0.15 on this Furuta model.
- Late bursts can strongly disrupt HOLD. ZERO keeps the terminal cost small on this plant.
- Window length changes the attack budget and its timing. Policy selection requires plant-specific worst-case and false-alarm checks.

Source: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025 · DOI 10.1145/3716550.3722013 · Fig. 4

Table 1: optimised attack sequences

Furuta HOLD; $\hat{p}_m = .25$, $\bar{a} = 4$, $\ell = 25$; binomial test

n_w	Attack	% in $\mathcal{W}$	$\sum_i (a_i)$
10	1110111000111000111011100	60%	15
12	1011110001101000111101100	58%	14
14	1101111000110001111011000	57%	14

Paper Table 1: optimal miss schedules for three binomial-window lengths.

Takeaways

- Each row reports the schedule, critical miss fraction in one window (% in $\mathcal{W}$), and total misses $\sum_i a_i$.
- The fraction is the critical count divided by n_w: $6/10 = 60\%$, $7/12 \approx 58\%$, $8/14 \approx 57\%$.
- **Link to our extension:** these attacks optimise against a fixed defence. We study how the defence configuration changes their worst-case cost.

Source: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025 · DOI 10.1145/3716550.3722013 · Table 1

(i) Impact isn't monotonic in ℓ

ω peaks around $\ell \approx 10$–13, depending on the panel; the optimiser concentrates or interleaves delays by initial conditions and budget.

(ii) Tight watchdogs help; loose ones add little

In Fig. 2, $\bar{a} = 1$–2 constrains the impact; from $\bar{a} = 3$ the curves barely change.

(iii) Geometric (χ^2) constraint binds

The geometric (χ^2) and combined panels coincide in Fig. 3. Compare detectors under equal session false-alarm budgets before ranking them.

(iv) ZERO beats HOLD on the Furuta

For the Furuta pendulum, zeroing the input on a miss is a far better defence; window length trades off per policy.

Source: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025 · DOI 10.1145/3716550.3722013

From the paper's gaps to our extension

What the paper shows

- Tight watchdogs ($\bar{a} \leq 2$) cut the impact; beyond $\bar{a} \approx 3$ the curves are flat (Fig. 2).
- A binding geometric (χ^2) constraint in Fig. 3.
- Strong policy sensitivity on Furuta in Fig. 4.
- Attack optimisation against a **fixed** defence in Table 1.

What we add

- Defender formulation: minimise $\max_\ell \omega^*$ subject to a session false-alarm limit.
- **Done:** independent reproduction, Furuta/tank policy replays, partial binomial sweep.
- **Planned:** geometric (χ^2) and combined-detector sweeps; missing $\ell = 25$ solves; empirical false-alarm estimates.

The paper asks how damaging an accepted attack can be. Our defender study asks which *tested* configuration lowers that cost at an acceptable false-alarm rate.

Source: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025 · DOI 10.1145/3716550.3722013

PRESENTATION PART 4

Setup & Reproduction

Paper Sections 3–5 and Appendix A.4.1

The authors' code, all related cases and reproduction of the reduced experiment

Presented by

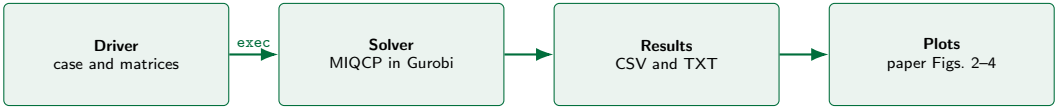
Abhishek Sharma

Our replay and analysis. Setting and code: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025

How the authors organised the code

Driver, solver, saved results and plots

A driver selects one configuration. The detector-specific solver searches for its worst accepted attack, then the plotting files read the saved costs.



The authors' REP pipeline: one optimisation for each selected configuration and horizon.

Drivers and model

`main.py`, `main_short.py`, `main_fig2/3/4.py`.

`define_system()` stores the closed-loop matrices; nested loops set the cases (paper Sections 3 and 5.1).

How settings reach the solver

`exec()` passes the loop settings as globals. Each figure driver changes the selected loop values.

Three solver scripts

`optimise_bin.py`
`optimise_chi2.py`
`optimise_bin_chi2.py`.

The combined script applies both statistical tests (paper Section 4, Eqs. 4–11).

Result and plot files

`results_{test}/{system}/`; filenames encode $\bar{a}$, $\hat{\rho}_m$, α, n_w. One row per saved length; TXT also stores schedules. `plot*.tex`, `fig/*.tex` and `Makefile` plot ω against ℓ.

Source: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025 · DOI 10.1145/3716550.3722013 · Authors' REP code; our code-structure explanation

Every case the code can run, and our focus

Three plant setups and three detector choices

Plant setup	Binomial	Geometric (χ^2)	Combined
`furuta_zero`: unstable; ZERO	Figs. 3, 4	Fig. 3	Figs. 2, 3; **our focus**
`furuta_hold`: unstable; HOLD	Fig. 4, Table 1	`main.py`	`main.py`
`qt_nmp`: stable; HOLD	Fig. 3	Fig. 3	Figs. 2, 3

Driver to output

`main.py`	all 9 cases; full sweep
`main_fig2.py`	Fig. 2; $\bar{a} = 1$–4, $n_w = 20$
`main_fig3.py`	Fig. 3; $\bar{a} = 4$, $n_w = 20$
`main_fig4.py`	Fig. 4/Table 1; $\bar{a} = 4$, $n_w = 10$–18
`main_short.py`	one Fig. 2 curve; $\bar{a} = 3$, $\hat{p}_m = .25$, $n_w = 20$

Sweep and code conventions

$\ell = 1$–25, $\bar{a} = 1$–5, $\hat{p}_m = .01, .03, \ldots, .25$, $n_w = 10, 12, \ldots, 20$, $\alpha = .01$. Figure curves use .09, .17, .25.

Window longer than sequence: one binomial check. Geometric-only has no window: run once at 20. Boundary slots are hits. Tank output is half its level: output cost is REP cost divided by four.

Our focus: the reduced case

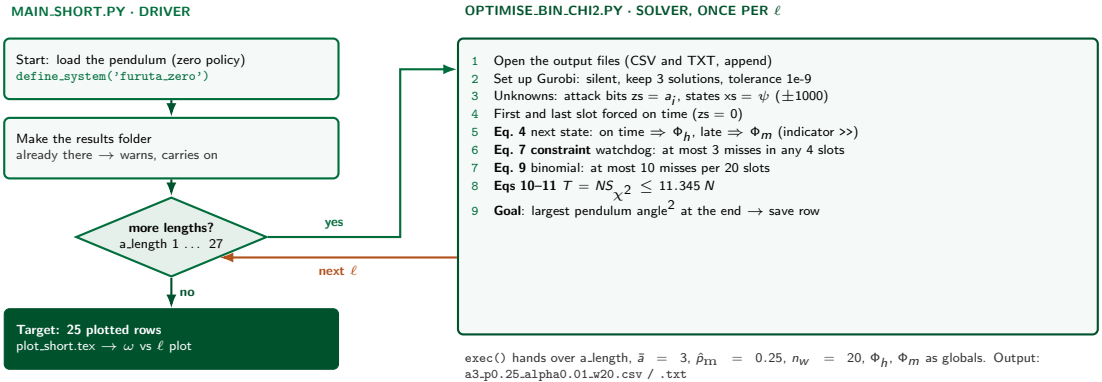
27 loop iterations: `a_length=1..27`; the intended plotted range has **25 positive-horizon rows**, $\ell = \texttt{a_length} - 2 = 1..25$. The authors estimate about 2 h for this reduced experiment; the full driver has 68,445 calls. Our independent enumeration uses the **same model and constraints**, rather than re-running Gurobi.

Source: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025 · DOI 10.1145/3716550.3722013 · REP drivers; Appendix A.4.1

How main_short.py runs, step by step

Part 4 · the code

The author driver loops over 27 values of `a_length`. Its plotted positive horizons are $\ell = 1, \ldots, 25$.



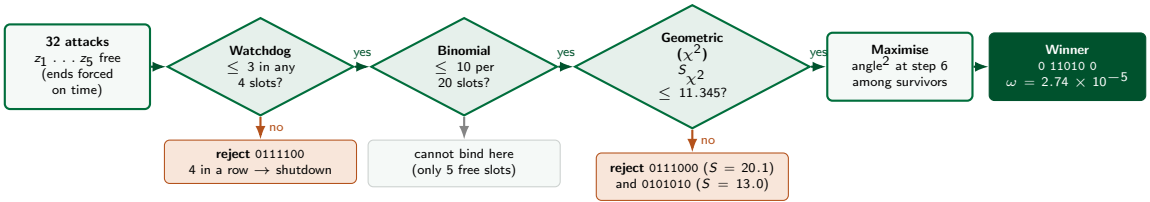
Author-driver control flow; output requires a successful solve, not merely a loop iteration.

Source: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025 · DOI 10.1145/3716550.3722013

Worked example: how the model selects an attack

Part 4 · the code

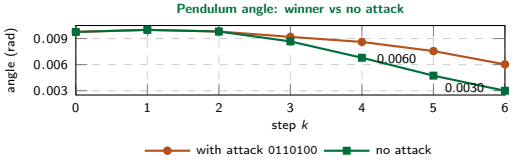
$\ell = 5$ (stored length $\ell + 2 = 7$): 32 possible attacks. Three checks remove the risky ones; the most damaging survivor wins.



$S_{\chi^2} = \sum_r (m_r - p_r N)^2/(p_r N)$. REP uses $T = N S_{\chi^2}$. For the winner, $N = 3$: $T = 19.0 \leq 11.345 \times 3 \approx 34.0$ ($\bar{a} = 3$, $\alpha = .01$).

Attack	Check	angle^2	Result
0000000	no attack	0.89	baseline
0111100	4 in a row	5.75	watchdog stop
0111000	$S = 20.1 > 11.345$	4.99	burst too rare
0101010	$S = 13.0 > 11.345$	2.22	too regular
0110100	$S = 6.33 \leq 11.345$	**3.63**	**chosen: hidden**

angle^2 at step 6, in units of 10^{-5}

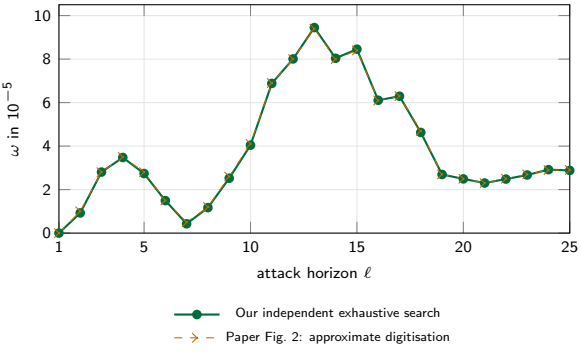


Candidate schedules and the same-initial-state angle replay; angle-squared costs use 10^{-5}.

Our replay and analysis. Setting and code: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025

Independent reproduction of the Furuta curve

Furuta, ZERO policy; combined tests; $\bar{a} = 3$, $\hat{p}_m = .25$, $n_w = 20$, $\alpha = .01$



Same axes; the paper series is a low-resolution graphical reference, not raw author data.

	Ours	Paper plot
Peak ω (10^{-5})	9.45	$\approx$ 9.4
Peak at ℓ	13	$\approx$ 13

Two checkable sequences

$\ell = 5$: 11010

$\ell = 15$: 101011011101000.

Our curve uses $\bar{a} = 3$; the matching sequences quoted in paper Section 5.4.1 use $\bar{a} = 4$. Neither needs more than three consecutive misses. The printed sequences omit REP's forced boundary hits.

Method and interpretation

Independent enumeration, about 6 min, with the authors' matrices and the same constraints. Short attacks exploit the initial tilt; the peak interleaves hits/misses; longer attacks wait and strike late.

The largest difference to the rounded graphical reference is below 0.08×10^{-5}; digitisation uncertainty is about 0.15×10^{-5}. This is visual agreement, not an exact raw-data error bound.

Our replay and analysis. Setting and code: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025 · Our enumeration; paper Fig. 2/Section 5.4.1; project links in glossary

PRESENTATION PART 5

Visualisation Cases and Defender Extension

Paper Section 5: Experimental Results

Results replay and validation · Project extension: defender configuration

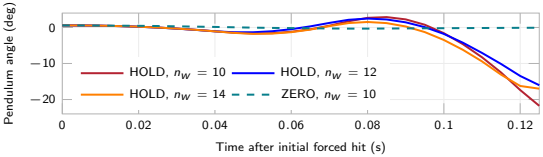
Presented by

Srinjoy Chakraborty

Our replay and analysis. Setting and code: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025

Furuta: the three published HOLD attacks

$\ell = 25,\ \hat{p}_m = 0.25,\ \bar{a} = 4,\ \alpha = 0.01,\ T_s = 5\,\mathrm{ms}$. The same schedule and initial state drive both policies.





Offline dashboard screenshot: schematic angle view.

All pass watchdog and binomial. The geometric (χ^2) test rejects all three: $S_{\chi^2} = 187.30,\ 86.22,\ 94.78 > 13.277$.

Textbook statistic; $\bar{a} = 4$, $\alpha = .01$, critical value $\chi^2_{.99}(4) = 13.277$.

Window	Misses	ω_{HOLD} (rad^2)	ω_{ZERO}, same attack	Peak $\lvert\theta\rvert$, HOLD
10	15	0.144120	-7.28×10^{-6}	21.75°
12	14	0.078222	-5.72×10^{-6}	16.03°
14	14	0.087986	-7.36×10^{-6}	17.00°

Observation: same-schedule replay supports the paper's actuation-policy sensitivity.

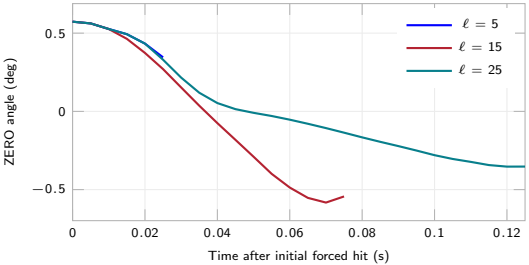
Our replay of paper Table 1 schedules: the attacks optimise HOLD under the binomial test.

$\omega < 0$: the schedule leaves a smaller terminal cost than nominal under ZERO. This does not bound the ZERO worst case; the attacker must be re-optimised for ZERO.

Our replay and analysis. Setting and code: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025 · Our replay of Table 1; project links in glossary

Furuta: horizon and detector validation

The published ZERO cases use the combined detector with $n_w = 20$. The extra 23-period replay uses binomial detection with $n_w = 10$.



Case	ω_{HOLD}	ω_{ZERO}
Paper, $\ell = 5$	-8.80×10^{-6}	2.74×10^{-5}
Paper, $\ell = 15$	1.18×10^{-4}	8.46×10^{-5}
Paper, $\ell = 25$	9.01×10^{-6}	2.88×10^{-5}
Extra, $\ell = 23$	4.24×10^{-2}	-4.93×10^{-6}

For the three published cases, $\chi^2 = 6.33,\ 12.83,\ 12.11$: all pass the combined constraints.

The extra replay has $\chi^2 = 126.11$. It passes binomial detection and fails the combined detector.

Observation: these cases show that impact depends on the schedule and active detector. A longer horizon does not imply larger terminal impact.

Separate attack problems; values use rad^2. The extra 23-period replay is feasible for binomial detection, with no proof of optimality.

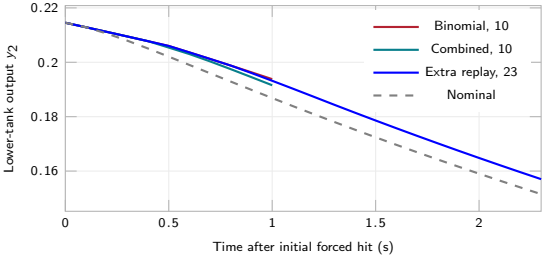
Published ZERO-optimised schedules at $\ell = 5, 15, 25$; HOLD values replay those same schedules.

Textbook S_{χ^2}; $\bar{a} = 4$, $\alpha = .01$, limit 13.277. Negative ω denotes a lower terminal cost than nominal under the other policy.

Our replay and analysis. Setting and code: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025 · Our replay; paper Section 5.4.1/Fig. 2; project links in glossary

Quadruple tank: detector comparison and replay

Paper HOLD model, $T_s = 0.1\,\mathrm{s}$, $\hat{\rho}_m = .25$, $\bar{a} = 4$, $\alpha = .01$. Two 10-period searches enumerate all 2^{10} schedules.



Case	HOLD ω	χ^2
Binomial, $\ell = 10$	0.003162	338.78
Combined, $\ell = 10$	0.002074	10.71
Extra, $\ell = 23$	0.001549	15.00

All pass their watchdog and binomial constraints. Only the combined-search case passes $\chi^2 \leq 13.277$.

Adding the geometric (χ^2) constraint reduces the 10-period maximum by **34.4%** for this configuration. This is a constraint comparison, not a ranking at equal session false-alarm rates.

Observation: this finite-horizon result supports the paper's detector comparison in Fig. 3.

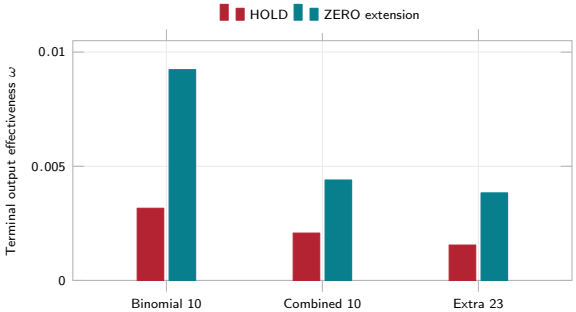
The curves show output deviations from equilibrium, not absolute water heights. The extra 23-period schedule is feasible for its binomial test, with no proof of optimality.

Our tank-output replay; $\ell = 10$ optima enumerate all 1,024 schedules, while the extra $\ell = 23$ case is not proven optimal.

Our replay and analysis. Setting and code: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025 · Our replay; paper Figs. 2–3; project links in glossary

Quadruple tank: policy comparison under a replayed schedule

Each pair uses the same schedule and initial state. HOLD is the paper model. Tank ZERO is a project extension.



All three tank schedules produce more terminal damage under ZERO. The increases are 2.92×, 2.12×, 2.48×, respectively.

Tank ZERO sets the next input *deviation* to zero. It does not mean switching the physical pumps off.

The paper's tank output is

$$y_k = \tfrac{1}{2}[x_{1,k}, x_{2,k}]^\top, \qquad \omega_y = \tfrac{1}{4}\omega_{\mathrm{REP}}.$$

Interpretation: the better actuation policy depends on the plant and tested attack. The Furuta comparison does not establish a universal ZERO rule.

What is needed to rank policies

These schedules were optimised for HOLD (or found for HOLD in the extra case). Re-optimise the attacker under ZERO before ranking policies by worst-case cost.

Same-schedule output-cost comparison; tank ZERO is our actuator-memory extension.

Our replay and analysis. Setting and code: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025 · Our replay with REP tank matrices and a ZERO extension; project links in glossary

Defender study: formulation and completion status

For each measured nominal scenario $\hat{\rho}_m$, choose the policy, detector, window and significance level:

$$d^*(\hat{\rho}_m) = \arg\min_{d} \max_{\ell \in \mathcal{L}} \omega^*(d, \hat{\rho}_m, \ell) \quad \text{subject to} \quad \mathrm{FA}(d, \hat{\rho}_m) \leq \varphi.$$

Quantity	Study settings	Status
Policy	Furuta HOLD / ZERO	Done: replay
Window n_w	10, 12, 14, 16, 18, 20	Partial
Significance α	.005, .01, .05	Partial
Scenario $\hat{\rho}_m$	.09, .17, .25; measured property	Per scenario
Watchdog $\bar{a}$	4	Fixed
Detector	binomial	Partial sweep
	geometric (χ^2), combined	Planned
Quick horizons	5, 10, 15, 20, 25	25 missing
Full horizons	1, . . . , 25	Planned
Session FA	nominal Bernoulli simulation	Planned

False-alarm constraint

Simulate many nominal Bernoulli($\hat{\rho}_m$) sessions of a declared length H_{sess}. Estimate FA as the fraction of sessions with at least one alarm; report uncertainty. Specify whether watchdog shutdown is included.

Interpreting the grid

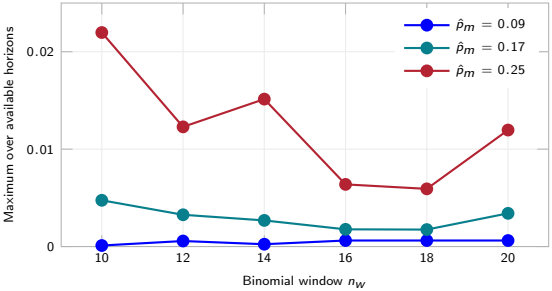
Current binomial design: 36 defender choices per scenario; 540 quick-grid or 2,700 full-grid calls across all three scenarios. The planned three-detector grid has 1,170 quick or 5,850 full calls if geometric-only windows are deduplicated. No completed cross-detector ranking is claimed.

$\hat{\rho}_m$ is measured, not chosen by defender or attacker. α is a per-test level, not session FA. Without an FA constraint, a damage-only search can prefer an impractically aggressive grid edge.

Our replay and analysis. Setting and code: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025 · Our defender formulation and partial binomial data; project links in glossary

Defender validation: horizon coverage changes the conclusion

Quick-grid sweep: HOLD, $\bar{a} = 4$, $\alpha = .01$, binomial test. The displayed maxima cover $\ell \in \{5, 10, 15, 20\}$; all $\ell = 25$ runs are still missing.



HOLD, $n_w = 10$, $\hat{p}_m = .25$, $\alpha = .01$:

Evidence	Effectiveness ω
Saved maximum, $\ell \in \{5, 10, 15, 20\}$	≈ 0.021973
Table 1 replay, $\ell = 25$	0.144120

The known 25-period attack is **about 6.56 times** the displayed maximum over available horizons.

Supported finding: window choice changes observed attack impact, and the trend is not monotone. This feasible attack gives a lower bound on worst-case damage, not a safety upper bound.

Remaining work: finish the missing solves and false-alarm checks before ranking the full grid. The saved log reports Gurobi error 10010: model too large for the size-limited licence, rather than a timeout.

Plot: HOLD, $\alpha = .01$, $\bar{a} = 4$, binomial detector. Values reproduce the notebook's rounded table; 432/540 runs saved across the full quick grid.

Partial binomial-sweep maxima by nominal scenario; the Table 1 replay exposes the missing-horizon effect.

Our replay and analysis. Setting and code: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025 · Our partial notebook sweep and Table 1 replay; project links in glossary

Conclusion

What the paper shows

- Deadline misses form an attack surface without forged data.
- Statistical stealth means that the active tests fail to reject the miss sequence.
- Impact depends on the plant, schedule and actuation policy. The paper demonstrates severe Furuta disruption under HOLD.
- MIQCP formulates the finite-horizon worst case under the chosen model and tests; a certified optimum requires solver bounds or exhaustive search.

What we did

- Independently enumerated the $\bar{a} = 3$ Furuta curve.
- Replayed Table 1: all three schedules fail the geometric (χ^2) test.
- Compared policies using the same schedules; tank ZERO can increase terminal cost.
- Formulated a defender min–max study and reported a partial binomial sweep.

Limitations

Terminal-cost objective; independent nominal misses; tested $\hat{\rho}_m \leq .25$; no project hardware experiment; missing $\ell = 25$ solves and session false-alarm estimates.

Source: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025 · DOI 10.1145/3716550.3722013

References

1. T. Nauta, H. Sandberg & M. Maggio. *Stealthy Computational Delay Attacks on Control Systems*. ACM/IEEE ICCPS 2025. DOI: 10.1145/3716550.3722013.

2. Authors' repeatability code, REP v1.2. Zenodo DOI: 10.5281/zenodo.14792914; github.com/Talitha-Nauta/REP/tree/v1.2.

3. A. M. H. Teixeira, I. Shames, H. Sandberg & K. H. Johansson. *A secure control framework for resource-limited adversaries*. Automatica 51 (2015), 135–148. DOI: 10.1016/j.automatica.2014.10.067. Taxonomy source: reference 38 in the delay-attack paper.

4. Our replay and defender-study materials: project links in the glossary. These remain explicitly labelled editable placeholders.

Source: Nauta, Sandberg & Maggio, *Stealthy Computational Delay Attacks on Control Systems*, ICCPS 2025 · DOI 10.1145/3716550.3722013